Yousef Ammar Abdullah

AI and ML Engineering | Arabic NLP
Makkah, Saudi Arabia | Phone | Email | Portfolio | GitHub

Summary

Data Science undergraduate and ArabicNLP 2026 co-author with experience building Arabic retrieval, OCR, analytics assistants, and classification APIs. Focused on model evaluation, explainability, and reliable tool integration.

Education

Bachelor of Data Science | Umm Al-Qura University, Makkah
2023 - 2027 (expected) | GPA: 3.92/4.00

Experience

AI Developer | Qatfah | Remote | 2026 - Present
AI Engineer, Part-time | Moasherat | Makkah | 2026 - Present

- Developed an Arabic analytics chatbot connecting FastMCP, Cube, Directus, and LibreChat to operational data; improved tool definitions and routing.
- Built a PaddleOCR and FastAPI pipeline to extract structured fields from Arabic Umrah permit images.

Data Scientist Intern | Qanoniah | Makkah | Feb - Aug 2026

- Built Arabic legal retrieval workflows with structured document parsing, hybrid search, cross-encoder reranking, and Hit@K/MRR evaluation.
- Developed Arabic OCR processing with text normalization and word/character error-rate evaluation; built a cached FastAPI leaderboard service.

AI Engineer Intern | Rakaya | 2026 - Aug 2026

- Enhanced a Laravel Arabic copilot with multi-round tool calling, retry handling, query caching, and clearer tool descriptions.

Data Analyst, Hajj Season | Moasherat | Makkah | May - Jun 2025

- Monitored operational KPIs and analyzed satisfaction-survey data for Hajj reporting.

Publication

Co-author | NAMAA Community at HalluScoring 2026: Beyond Predictive Performance: Reliability and Generalization in Arabic Hallucination Detection.
ArabicNLP 2026 Shared Tasks, pp. 1138-1145. Team result: 3rd of 7 teams; hidden-test ROC-AUC 0.7596. Publication

Selected Projects

Arabic Complaint Classification API | Python, MARBERT-v2, FastAPI, Docker

- Integrated sentiment, topic, and intent classifiers with deterministic action routing, confidence-based manual review, and optional LLM explanations.

Agents World | JavaScript, Node.js, Claude Code, Codex

- Built a local interface for viewing and continuing AI coding sessions, with access controls, recoverable session deletion, and agent activity tracking.

Arabic Legal RAG API | FastAPI, Qdrant, E5 embeddings

- Built retrieval and cited-answer generation over legal documents, with a context-only fallback when no language model is configured.

Interpretable Cardiology Risk | scikit-learn, SHAP, PyMC

- Compared logistic regression and random forest models on Framingham data, using recall, ROC-AUC, Bayesian analysis, and SHAP explanations.

Technical Skills

Python, SQL, JavaScript, PHP/Laravel; PyTorch, Hugging Face Transformers, scikit-learn; FastAPI, FastMCP, Docker, Git; Qdrant, ChromaDB, BM25, reranking; PaddleOCR, SHAP, PyMC.
Languages: Arabic (native); English (professional working proficiency).